

```
import java.io.*; import java.util.*;
```

```
class S implements Serializable{  
    String n; Map<String,Integer> m=new LinkedHashMap<>();  
    S(String n){this.n=n;}  
    double a(){ if(m.isEmpty()) return 0;  
        int s=0; for(int v:m.values()) s+=v; return s/(double)m.size(); }  
}
```

```
public class G{  
    static final String F="students.ser";  
    List<S> a=new ArrayList<>(); Scanner sc=new Scanner(System.in);  
    public static void main(String[] args){ new G().go(); }  
  
    void go(){ load();  
        for(;;){  
            System.out.print("\n1.L 2.A 3.S 4.R 0.X > ");  
            switch(sc.nextLine().trim()){  
                case "1" -> list();  
                case "2" -> add();  
                case "3" -> score();  
                case "4" -> rank();  
                case "0" -> { save(); return; }  
                default -> System.out.println("?");  
            }  
        }  
    }  
}
```

```
S f(String n){ for(S s:a) if(s.n.equalsIgnoreCase(n)) return s; return null; }
```

```
void add(){  
    System.out.print("Name: "); String n=sc.nextLine().trim();  
    if(n.isEmpty()||f(n)!=null){ System.out.println("No/dup"); return; }  
    a.add(new S(n)); System.out.println("OK");  
}
```

```
void score(){  
    System.out.print("Name: "); S s=f(sc.nextLine().trim());  
    if(s==null){ System.out.println("NF"); return; }  
    System.out.print("Sub: "); String sub=sc.nextLine().trim();  
    System.out.print("Score: ");  
    try{ s.m.put(sub,Integer.parseInt(sc.nextLine().trim())); }  
    catch(Exception e){ System.out.println("Bad"); }  
}
```

```
void list(){  
    if(a.isEmpty()){ System.out.println("(empty)"); return; }  
    for(S s:a){  
        System.out.printf("%s(%.2f)\n", s.n, s.a());  
        for(var e:s.m.entrySet())  
            System.out.println(" "+e.getKey()+" "+e.getValue());  
    }  
}
```